Pseudo Code for Particle swarm Optimization for function optimization.

```
Begin: #include <stdio.h>
       #include <stdlib.h>
       #include <math.h>

#define N 30
#define MAX_ITER 100
#define W 0.7
#define C1 1.5
#define C2 1.5

#define XMIN -10
#define XMAX 10

double fitness(double x)
{
    return x * x;
}

int main()
{
    double X[N], V[N];
    double Pbest[N], PbestFitness[N];
    double Gbest, GbestFitness;
    double r1, r2;
    int i, iter;
    srand(time(NULL));
    srand(NULL);
```

```
for (i=0; i<N; i++)
{
    X[i] = XMIN + ((double) rand() / RAND_MAX) * (XMAX
                - XMIN);

    V[i] = 0;
    pbest[i] = X[i];
    pbestFitness[i] = Fitness(X[i]);
}
gbest = pbest[0]
gbestFitness = pbestFitnes[0];

for(i=1; i<N; i++)
{
    if (PbestFitness[i] < GbestFitness)
    {
        Gbest = Pbest[i];
        GbestFitness = PbestFitness[i];
    }
}

for (iter=0; iter<Max_ITER; iter++)
{
    for (i=0; i<N; i++)
    {
        r1 = random01();
        r2 = random02();

        V[i] = W * V[i]
             + C1 * r1 * (Pbest[i] - X[i])
             + C2 * r2 * (Gbest - X[i]);
```

Scanned with OKEN Scanner

```
X[i]=X[i]+V[i]

if (X[i] < XMIN)
    X[i]=XMIN;

if (X[i] > XMAX)
  X[i]=MAX;

double current= fitness(X[i]);

if (current < Pbest Fitness[i])

  {
   Pbest [i]= X[i];
    PbestFitness [i]= current;

  }

if (current < Gbest Fitness)

  {
   Gbest = X[i];
   Gbest Ficness = current;

  }
 }

Printf (Iterations:"
Printf (optimal solution)
Print f (g best)
Print f (Minimal solution)
return 0;
}
```

Neural N/w

Scanned with OKEN Scanner

Optimizing Tumor Classification Through Transfer Learning & Particle Swarm optimization Driven Fea - re Extraction (2024).

2) Problem Objective

The main objective is to improve neur performance using particle Swarm Op
- optimize the weights of the inception V Network
- Improve brain tumor classification accura
- Reduce training errors & improve model ge - zation
- combine PSO with the Adam optimizer to a - eve the better performance.



Code Summary :→

— x —

1) Generate a Classification dataset and split it int training & testing data

2) Create a neural network with input, hidden & output layers.

3) Initialize particles with random weights & vel - cities

4) Apply PSO using personal best & global be to optimize neural network.

5) Evaluate the optimized neural network & Calculate its accuracy

Scanned with OKEN Scanner

1) Title of the Paper:-

Optimizing Tumor Classification Through Transfer Learning & Particle Swarm optimization Driven Feature Extraction (2024).

2) Problem Objective

The main objective is to improve neural network performance using particle Swarm Optimization (PSO).

- optimize the weights of the inception V3 neural Network
- Improve brain tumor Classification accuracy
- Reduce training errors & improve model generalization
- combine PSO with the Adam optimizer to achieve the better performance.

Code Summary :→

1) Generate a Classification dataset and split it into training & testing data
2) Create a neural network with input, hidden & output layers.
3) Initialize particles with random weights & velocities
4) Apply PSO using personal best & global best to optimize neural network.
5) Evaluate the optimized neural network & Calculate its accuracy.

Scanned with OKEN Scanner

Output Summary:

1) Display the best fitness (loss) value obtained thro - ugh PSO

2) Shows the training accuracy of the neural network

3) Shows the testing accuracy of the neural network

4) Demonstrate optimization of neural network weights of PSO

5) Helps improve classification performance by finding better network weights

completed
CO5
2/11/26

Scanned with OKEN Scanner